SAPPhiRE|Q

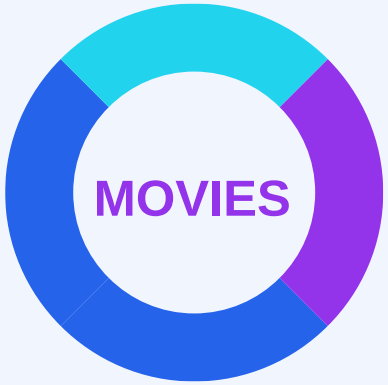
FRONT-END DEVELOPMENT • STUDENT PROJECT

Movie Explorer

React + React Router + TMDB API + Tailwind CSS

A step-by-step project for building a movie discovery app with search, detail pages and a favourites list, covering API integration, routing, debouncing and infinite scroll. Duration: 2 weeks.

4	React	TMDB	Router
Modules	Framework	Public API	Navigation



React • Router • TMDB

01 / PROJECT OVERVIEW

Movie Explorer

A movie browsing app powered by a public API.

Project Description

Students build a movie discovery app using the free TMDB API. Users browse trending and popular movies, search by title, open a detail page with cast and trailer, and save favourites that stay after a refresh.

Project Scope

Build a home page with movie rows, a debounced search, a movie detail page with routing, infinite scroll, a favourites page saved in localStorage, loading and error states, and a responsive Tailwind layout.

Prerequisites

- **React Basics:** Components, props, useState and useEffect.
- **JavaScript:** fetch, promises and async/await.
- **CSS Basics:** Flexbox and grid.
- **Setup:** Node 18+ and a free TMDB API key.

Target audience: Beginner to intermediate frontend students ready to build a multi-page app with a real API.

02 / PROJECT ROADMAP

Four-Module Movie Explorer Roadmap (2 Weeks)

Week 1: modules 1-2. Week 2: modules 3-4.

#	MODULE	CORE FOCUS	KEY TECHNOLOGIES
1	**Setup, API Key & Movie Rows**	Foundation	Vite, fetch, useEffect, .env
2	**Search with Debounce**	User Input	Controlled input, debounce, custom hook
3	**Routing & Movie Detail Page**	Navigation	React Router, URL params
4	**Favourites, Infinite Scroll & Polish**	Advanced UI	Context, localStorage, IntersectionObserver

Student Workflow

Complete each module in order, test in the browser after each step, and finish the exercise before moving on.

02 / MODULE 1

Setup, API Key & Movie Rows

Build the Movie Explorer step by step using React, React Router, the TMDB API and Tailwind CSS.

Learning Objectives

- Create the React project with Tailwind.
- Keep the API key in an .env file.
- Fetch and show trending and popular movies.

Core Concepts

- Environment Variables: the key is read from import.meta.env.
- Data Fetching: calling the API inside useEffect.
- Reusable Components: MovieCard and MovieRow.

Tools & Technologies

- Vite
- Tailwind CSS
- fetch
- TMDB API

Hands-on Student Exercises

Show two rows, Trending and Popular, each as a list of movie cards with poster, title and rating.

03 / MODULE 2

Search with Debounce

Build the Movie Explorer step by step using React, React Router, the TMDB API and Tailwind CSS.

Learning Objectives

- Build a search box.
- Wait until the user stops typing before calling the API.
- Show results, empty and error states.

Core Concepts

- Debounce: delaying the request to avoid too many calls.
- Custom Hook: a reusable useDebounce hook.
- Request States: loading, error and no results.

Tools & Technologies

- useState
- useEffect
- Custom hooks

Hands-on Student Exercises

Type a movie name, confirm only one request is sent after you stop typing, and show a message when nothing is found.

04 / MODULE 3

Routing & Movie Detail Page

Build the Movie Explorer step by step using React, React Router, the TMDB API and Tailwind CSS.

Learning Objectives

- Add routes for home, search and movie details.
- Open a detail page from a movie card.
- Show overview, cast and trailer.

Core Concepts

- Route Params: reading the movie ID from the URL.
- Nested Data: fetching details, cast and videos.
- Navigation: links, back button and a not-found page.

Tools & Technologies

- React Router
- useParams
- Embedded video

Hands-on Student Exercises

Click a movie to open /movie/:id and show the poster, overview, cast and a trailer.

05 / MODULE 4

Favourites, Infinite Scroll & Polish

Build the Movie Explorer step by step using React, React Router, the TMDB API and Tailwind CSS.

Learning Objectives

- Add and remove favourites.
- Load more movies when the user scrolls.
- Make the layout responsive and deploy.

Core Concepts

- Persistence: saving favourites in localStorage.
- IntersectionObserver: loading the next page near the bottom.
- Responsive Grid: card layout from phone to desktop.

Tools & Technologies

- Context API
- localStorage
- IntersectionObserver
- Vercel

Hands-on Student Exercises

Favourite three movies, refresh to confirm they stay, scroll to load more results, and deploy to Vercel.

06 / PROJECT OUTCOME

What You Will Build

The completed project combines four modules into a working movie discovery app.

Application Structure

Pages

Home, search, detail, favourites

Hooks & Context

Debounce and favourites store

TMDB API

Movies, search, cast, trailers

Pages call the TMDB API through hooks, results render as cards, and favourites are stored in context and localStorage.

Skills Practiced

- Fetching and displaying public API data.
- Debounced search with a custom hook.
- Routing with URL parameters.
- Infinite scroll with IntersectionObserver.
- State persistence with localStorage.

Use the module exercises as checkpoints while building your application.